

COMP6237 – Logistic Regression

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Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/>

(Thanks to Jason Noble and Cosma Shalizi whose lecture materials I used to prepare)

COMP6237: Logistic Regression

.Outline:

- Introduction
- Basic ideas of logistic regression
- Logistic regression using R
- Some underlying maths and MLE
- The multinomial case
- How to deal with non-linear data
 - Model reduction and AIC
- How to deal with dependent data
- Summary
- Problems

COMP6237: Logistic Regression

Learning Outcomes

- Understanding of logistic regression
- How to perform logistic regression using R
- How to deal with non-linear data
- How to deal with dependent data

Introduction

.Previous lecture: Linear regression

- tried to predict a continuous variable from variation in another continuous variable (e.g., basketball ability from height)

.Here: Logistic regression

- Try to predict results of a binary (or categorical) outcome variable Y from a predictor variable X
- This is a classification problem: classify X as belonging to one of two classes
- Occurs quite often in science ... e.g., medical trials (will a patient live or die dependent on medication?)

Dependent variable Y

Predictor Variables X



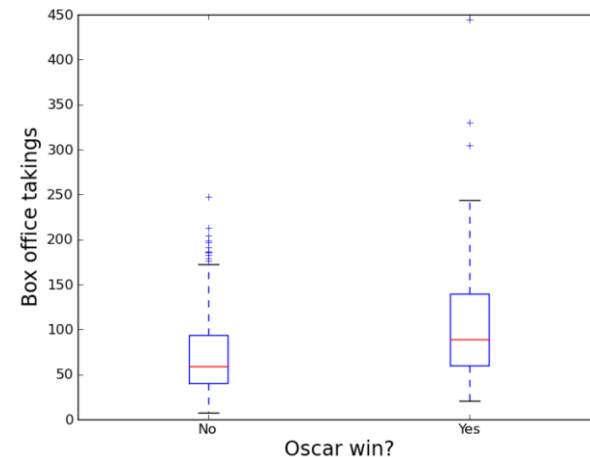
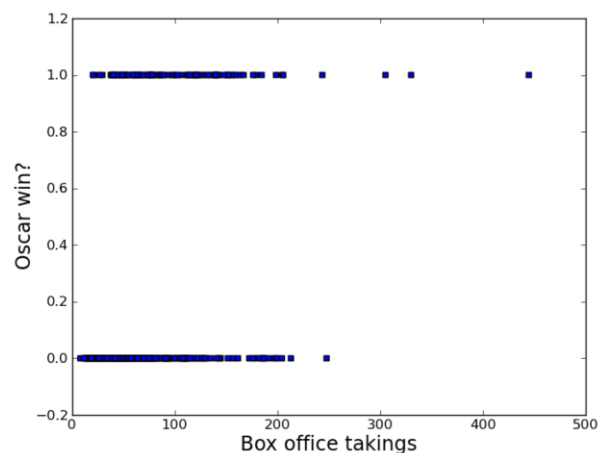
Fate	Treatment	Age (2005)
alive	placebo	45
dead	drug A	61
dead	placebo	29
alive	drug B	33
dead	placebo	70
dead	drug A	61
dead	placebo	44
alive	drug B	50

The Oscars Example

- A fictional data set that looks at what it takes for a movie to win an Oscar
- Outcome variable: Oscar win, yes or no?
- Predictor variables:
 - Box office takings in millions of dollars
 - Budget in millions of dollars
 - Country of origin: US, UK, Europe, India, other
 - Critical reception (scores 0 ... 100)
 - Length of film in minutes
 - This (fictitious) data set is available here:
 - <https://www.southampton.ac.uk/~mb1a10/stats/filmData.txt>

Predicting Oscar Success

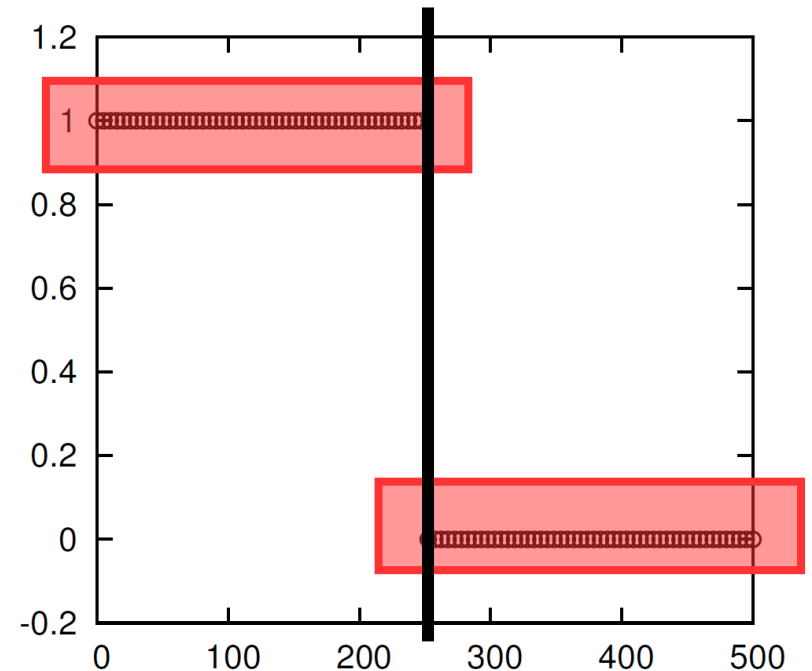
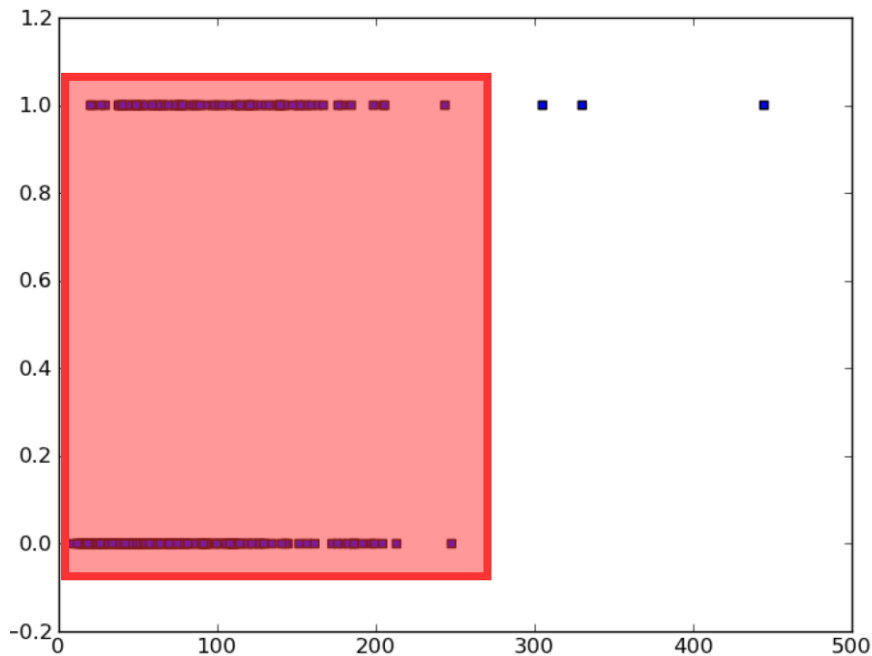
- Let's start simple and look at only one of the predictor variables
- Do big box office takings make Oscar success more likely?
- Could use same techniques as below to look at budget size, film length, etc.



Introduction (1)

.Could use a linear classifier ...

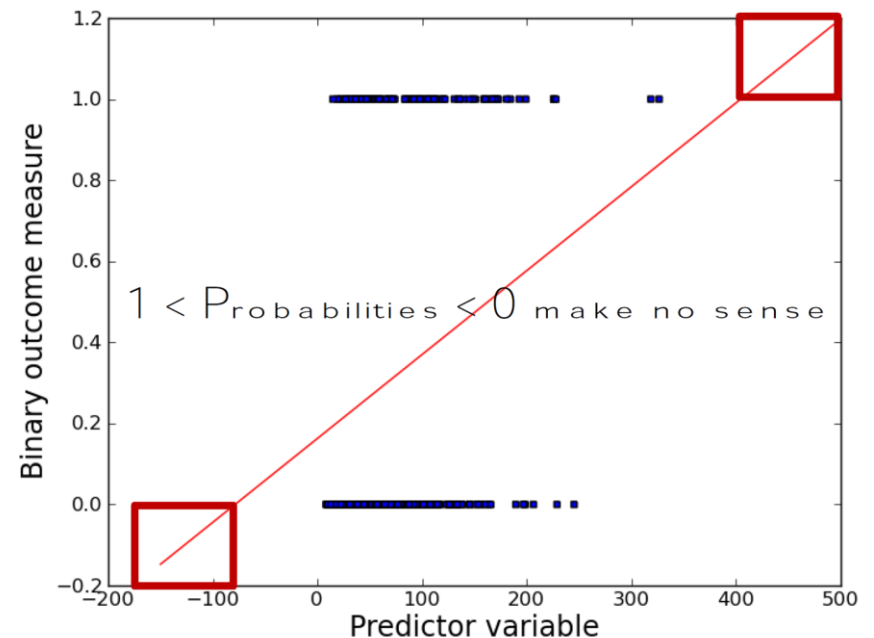
- But this does not give us probabilities, which are desirable if
 - . We want to handle different error costs between classes
 - . We need some indication of confidence
 - . Perfect classification is not possible



Introduction (2)

.Naive approach:

- Code binary variable as 0 or 1, do linear regression and interpret outcomes as probabilities ...
- Avoid range problem by assuming
 - $P(x)=0 \ x < 0$
 - $P(x)=1 \ x > 1$?
- Problems with saturation: once we reach boundaries, we cannot discriminate any more, model becomes insensitive to predictor



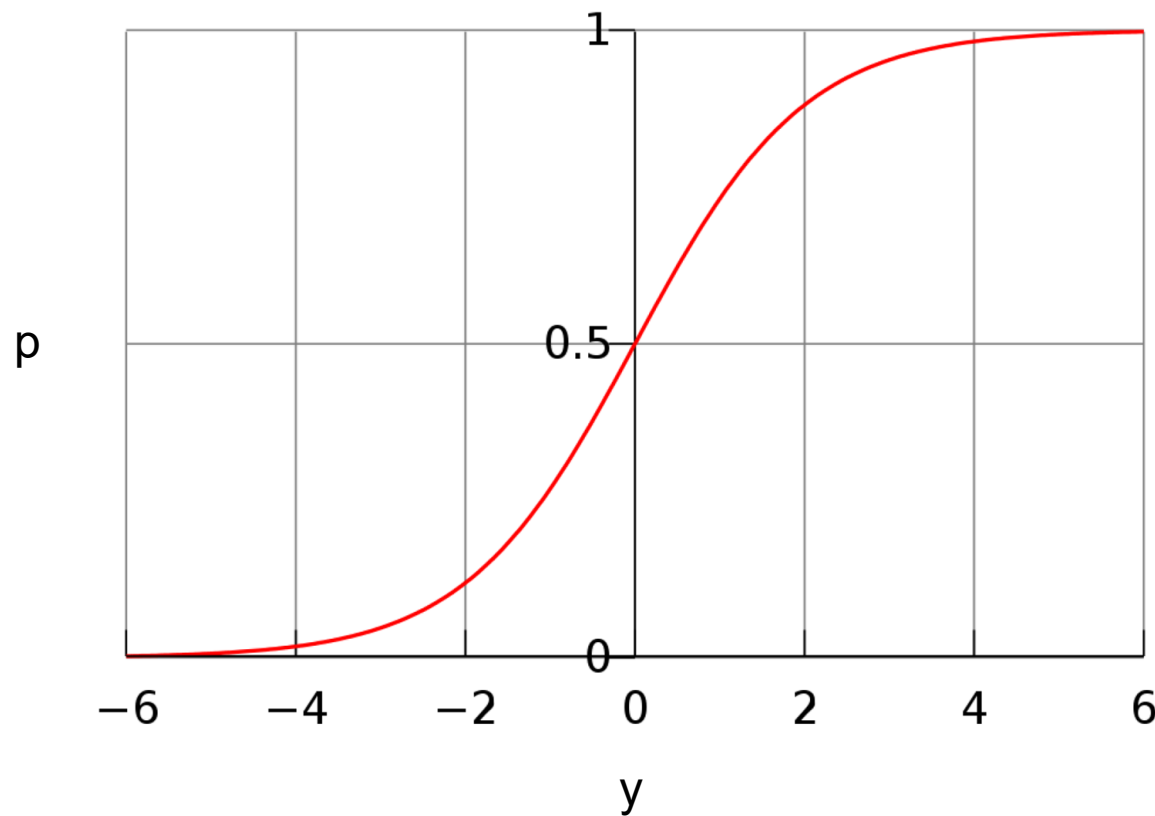
The Idea

- Can transform predictor variable to something we can do linear regression on!
- Want to find a probability $\Pr(Y=1|X=x)=p(x)$ for Y to be in class 1 (or 0)
 - Cannot do linear regression on p directly because of range issues
 - What about doing regression with $\log p(x)$ linear in x ? $\rightarrow$ log's only unbounded in one direction (whereas linear functions are not)
 - Easiest modification of $\log p$ that has an unbounded range is the logistic transformation $\log p(x)/(1 - p(x))$
 - $p/(1-p)$ is also called “odds”

An Aside: From Probability to Odds

- Odds are often used in gambling/betting
 - Odds = “probability of event”/”probability of not event”
- E.g.:
 - “9 to 1 against” $\rightarrow p=0.1$
 - “even odds” $\rightarrow p=0.5$
 - “3 to 1 on” $\rightarrow p=0.75$
- Not scientific parlance, don't write your work up like this.

The Logistic Function



$$y = \log \frac{p}{(1-p)}$$

Is the easiest transform to solve our range problems.

Logistic Regression

Formally, the logistic regression model is

$$\log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} = b + \mathbf{w}\mathbf{x}$$

$$p(\mathbf{x}) = \frac{1}{1 + \exp(-(b + \mathbf{w}\mathbf{x}))}$$

How to Run Logistic Regression in R

.Will use the Oscar example here, variables of interest are Oscar and BoxOffice

.Build the regression model in R and then use summary command to see information:

```
boxOfficeModel = glm( Oscar ~ BoxOffice,  
family=binomial (link="logit"))
```

```
summary(boxOfficeModel)
```

family is binomial, will understand
this later in the lecture

generalised linear model

R Output

p value; likelihood for the data to arise if there
Was no relationship between the variables

```
Deviance Residuals:
```

Min	1Q	Median	3Q	Max
-1.6432	-0.8316	-0.6997	1.2380	1.8546

```
Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.750349	0.256883	-6.814	9.50e-12	***
BoxOffice	0.011306	0.002507	4.510	6.48e-06	***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Logit increment per \$million in Box Office takings

logit score for a film that made \$0

Making Sense of the R Output

.What is the chance of a film with Box Office takings of \$50 million to win an Oscar?

- Logit score = $-1.75 + 0.011 * 50 = -1.2$

- $p = 1 / (1 + \exp(-\text{logit})) = 0.23$

- $\rightarrow$ the model predicts that such a film has a 23% chance to win an Oscar

Using the Other Variables

- What about the other predictor variables?
 - Could look at each variable separately with a logistic regression model and check if it has any explanatory value ...
 - Better include them jointly to fit all the variables in the same model:

```
fullModel = glm( Oscar ~ BoxOffice + Budget + Country +  
Critics + Length, family=binomial (link="logit"))
```

- (could handle this for normal linear regression in the same way)

Review of our Strategy

Original
 Y



0



1



Y as a
probability



0



1



Transformed

$g(Y) \in (-\infty, +\infty)$



$-\infty$



$+\infty$



Linear model for g $g(Y) = mX + b$

Comments

• Logistic regression is a modelling choice, posit it, then check whether it works or has systematic flaws

• Reasons for using it:

- Tradition
- Often works surprisingly well as a classifier (But many simple techniques do that ...)
- Closely related to exponential family distributions (which e.g. arise out of maxent)
- Is problematic if data can be linearly separated (if b, w perfectly separates linearly, so does cb, cw with $c > 0$; so there is no parameter vector that maximises likelihood)

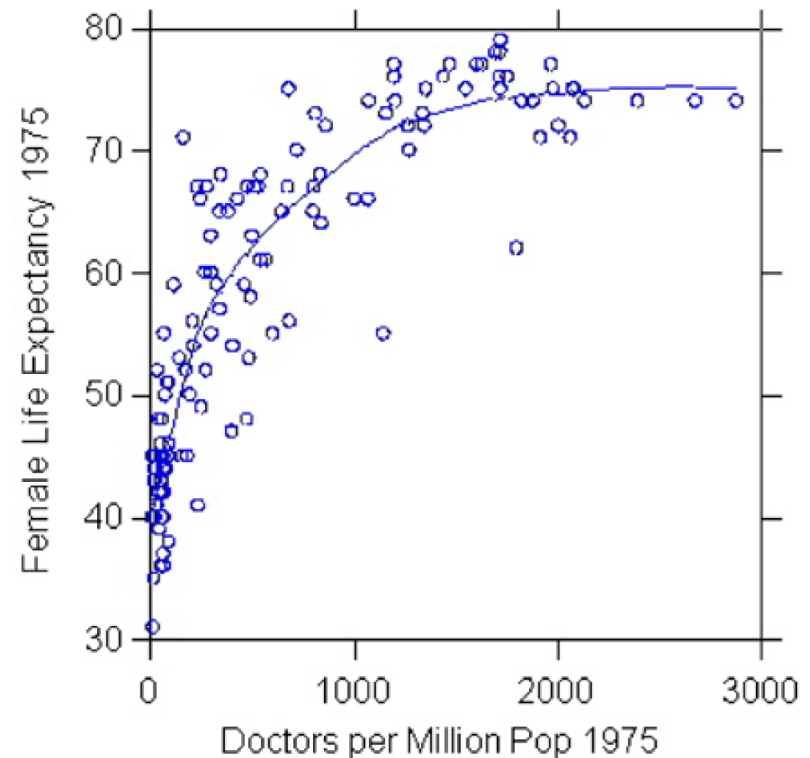
Non-Linear Data

Problems with Linear Regression

- Linear regression assumes a linear relationship between outcome variables and predictors; can we deal with non-linear data like those? →

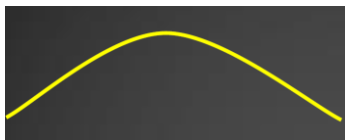
- (→ transformations of variables,
• fitting polynomials)

- Independence assumption, linear regression assumes that effects of predictors are independent of each other (→ interaction terms)

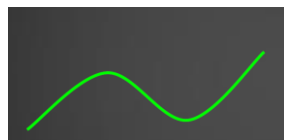


Non-linear Data: Fitting Polynomials

- Fit a polynomial instead of a straight line
- Same idea as linear regression, just turning one predictor (X) into several ($X^2, X^3, \dots$)
- Allows to deal with obvious non-linearity without having to specify in advance what transformation is
- Intuition for degree of polynomial should come from shape of relationship on scatterplot



parabolic
(2nd order)



cubic
(3rd order)



quartic
(4th order)

R practicalities

•1st way:

```
model = lm(y~x + I(x^2))
```

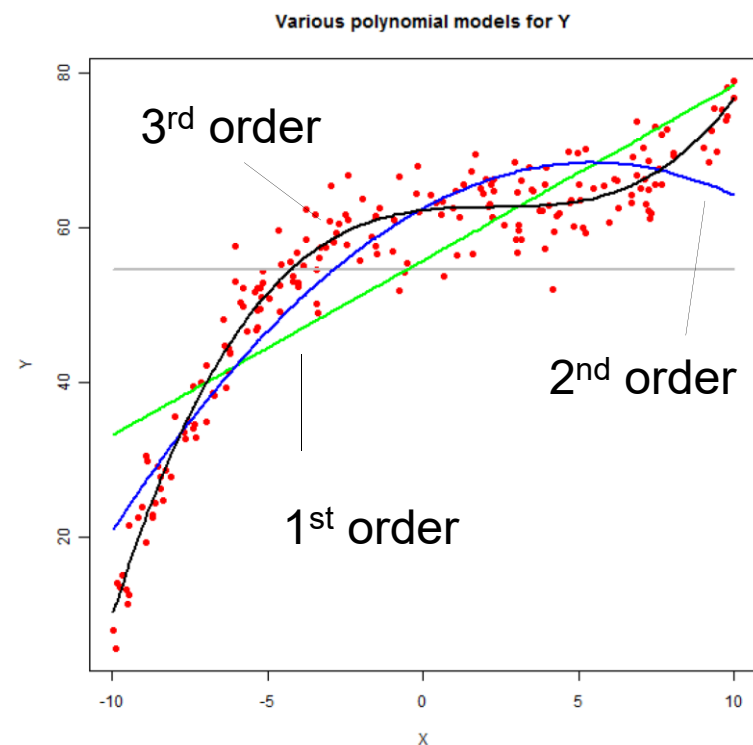
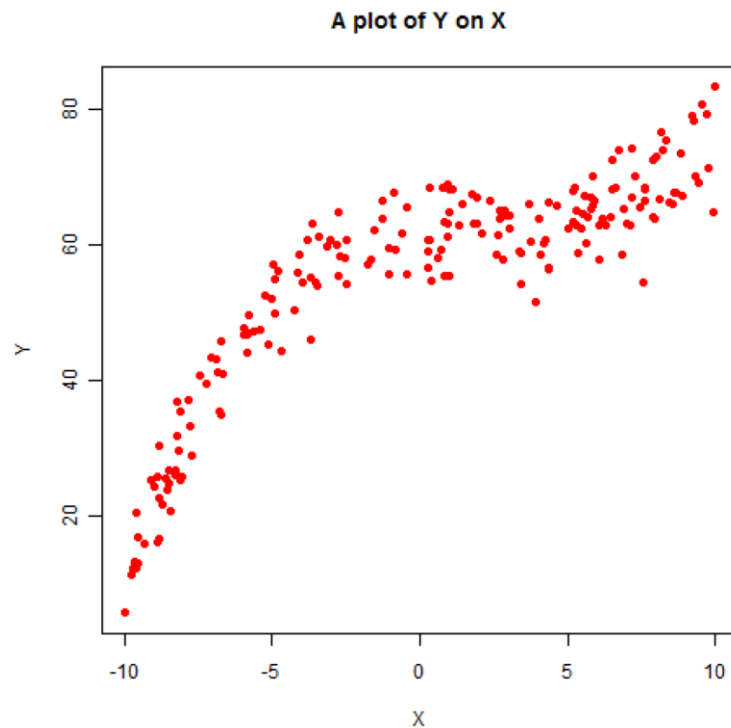
•(easy to interpret)

•2nd way:

```
model = lm(y ~ poly(x, 2))
```

- Uses orthogonal polynomials which are more efficient for fitting, but resulting coefficients not straightforward to interpret

Example



```
m0 = lm ( Y ~ 1 )  
m1 = lm ( Y ~ X )  
m2 = lm ( Y ~ X + I (X^2) )  
m3 = lm ( Y ~ X + I (X^2) + I (X^3) )  
m4 = lm ( Y ~ X + I (X^2) + I (X^3) + I (X^4) )
```

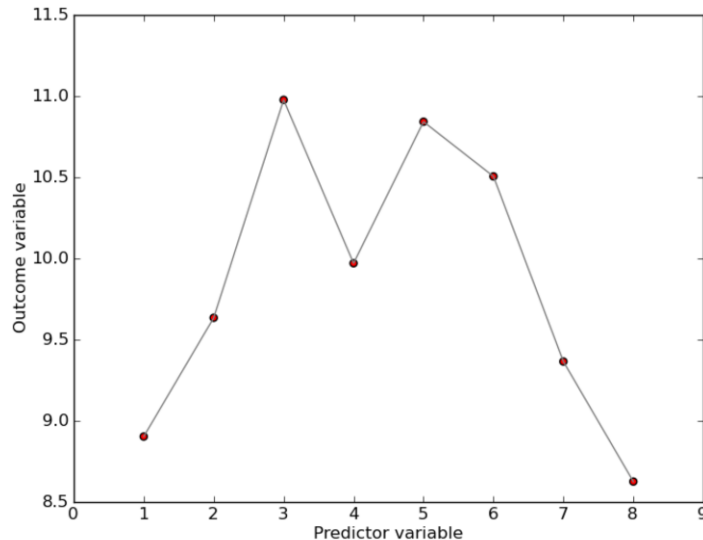
Problem: Higher order models will always do better ...end up with matching number of data points?

Occam's Razor

- William of Occam (1288-1348)
- All else being equal, the simplest explanation is the best one.
- In statistics this means:
 - A model with fewer parameters is to be preferred to one with more parameters.
 - ... but this needs to be weighed against a model's ability to predict.



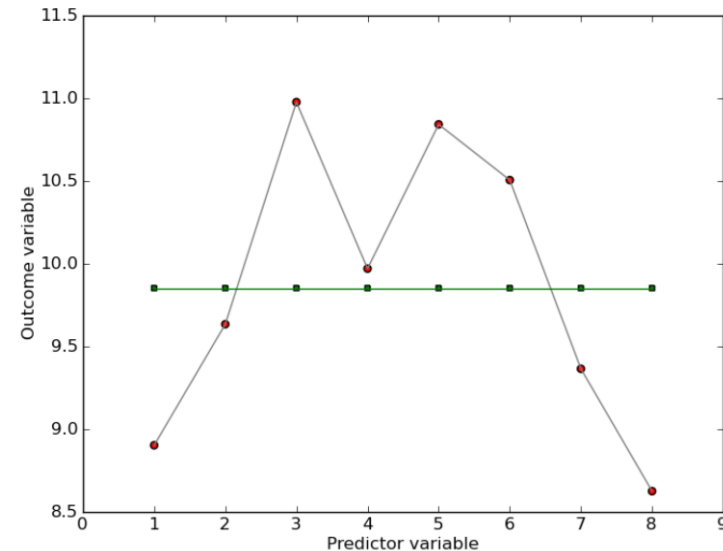
.Over-fitting



.N-1 predictors enough to replicate data. But this is an absurd model without explanatory power.

.(if you ask me the height of somebody I ask for information about his gender, DOB, occupation, address, parents, neighbours, friends, ...)

.Under-fitting



.Lowest number of predictors we can use is to just explain all the variation by the mean.

.(no variation explained, if you ask me the height of somebody in the UK, I say 1.67m ...)

The Machine learning way – Regularization

.Two reasons to do this:

- Problem ill-posed (more variables than observations)
- Solution does not generalize well

.Redefine optimization problem: add regularization term to residuals

$$E = \sum_{i=1}^N (y_i - \tilde{\mathbf{w}}\tilde{\mathbf{x}}_i)^2 = \|\mathbf{y} - \tilde{X}\tilde{\mathbf{w}}\|^2 \rightarrow \|\mathbf{y} - \tilde{X}\tilde{\mathbf{w}}\|^2 + \lambda\|\tilde{\mathbf{w}}\|_{1/2}$$

.Popular forms of norm:

- Ridge (Tikhonov): L^2 norm (sum squares)
 - $\rightarrow$ force some elements of w to be small
- Lasso: L^1 norm (sum absolute values)
 - $\rightarrow$ typically some elements of w can be forced to zero

Stats Methods

The old Ways of doing it

- Do model reduction through a series of F-tests, asking if models with more parameters explain significantly more variation in the outcome variable.

- → “Step-wise” model reduction.

- Either start with full model and reduce or start with simplest model and build up complexity.

- Caveats:

- Complicated ...

- Model one ended up could depend on the starting point ...

A better Way?

- Kullback-Leibler divergence:

- Measure of the informational distance between two probability distributions
- K-L distance between a real-world distribution and a model distribution tells us how much information is lost if we describe the real-world distribution with the model distribution.
- A good idea to obtain a good model is to minimise the K-L distance to the real-world.

Akaike's Information Criterion

- If we had a true distribution F and two models $G1$ and $G2$ we could figure out which model we prefer by calculating K-L distances $F-G1$ and $F-G2$; don't know F in real world cases, but can estimate $F-G1$ and $F-G2$ from the data
- AIC is an estimator for the K-L divergence
- Akaike's information criterion:

$$AIC = 2K - 2\log(L)$$

AIC score: the lower the better.

Number of predictors
– punishes models with many predictors

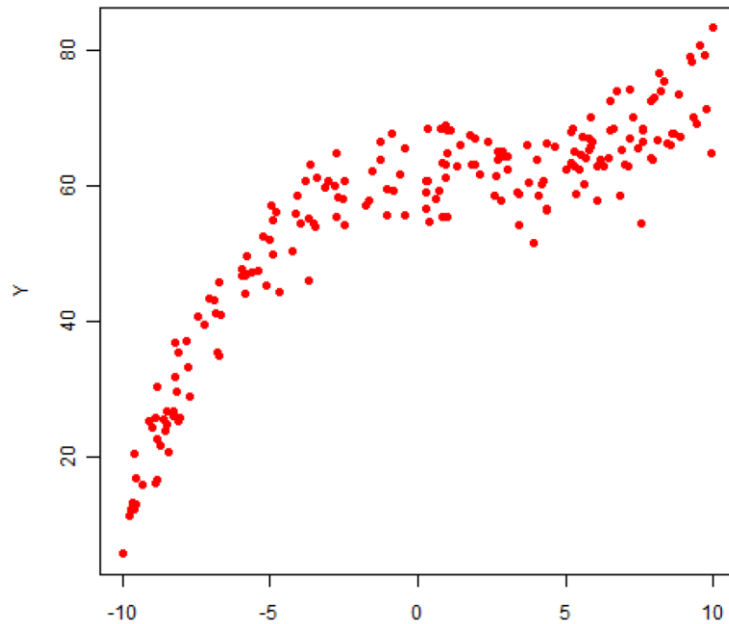
Maximised likelihood value
(using these predictors)
– rewards fit of model to data

R practicalities

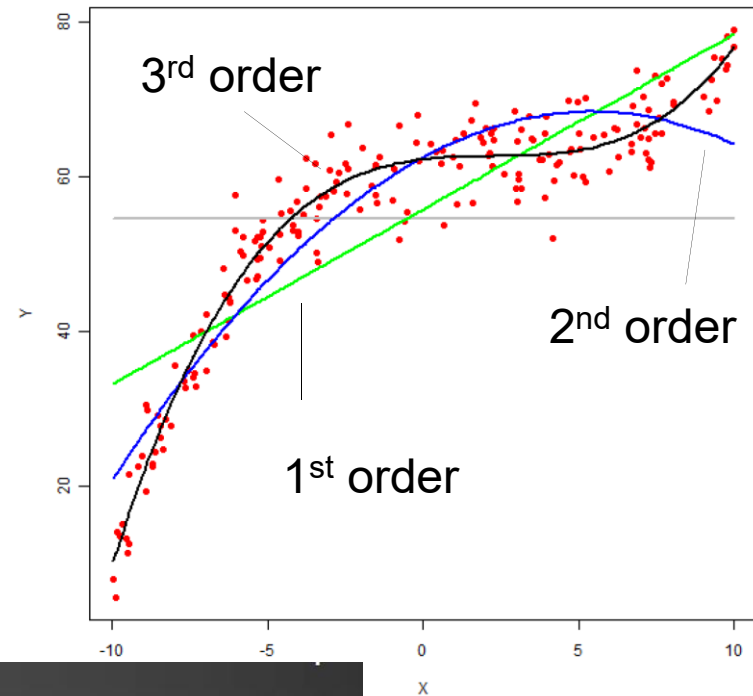
- Very easy to use in R. Suppose we have (regression) models $m_1, m_2, m_3, \dots$
- invoke: `AIC (m1,m2,m3,...)`
 - This will list AIC values of all the models, simply pick the lowest AIC score.
- `drop1 (model)`
 - Is also quite useful. It returns AIC scores for dropping each predictor in turn.

Back to the Example

A plot of Y on X



Various polynomial models for Y



```
m0 = lm ( Y ~ 1 )  
m1 = lm ( Y ~ X )  
m2 = lm ( Y ~ X + I(X^2) )  
m3 = lm ( Y ~ X + I(X^2) + I(X^3) )  
m4 = lm ( Y ~ X + I(X^2) + I(X^4) )
```

AIC (m0,m1,m2,m3,m4)

	df	AIC
m0	2	1691.918
m1	3	1435.907
m2	4	1278.603
m3	5	1110.864
m4	6	1111.103

← m3, the cubic model, has the lowest AIC score

Limitations of AIC

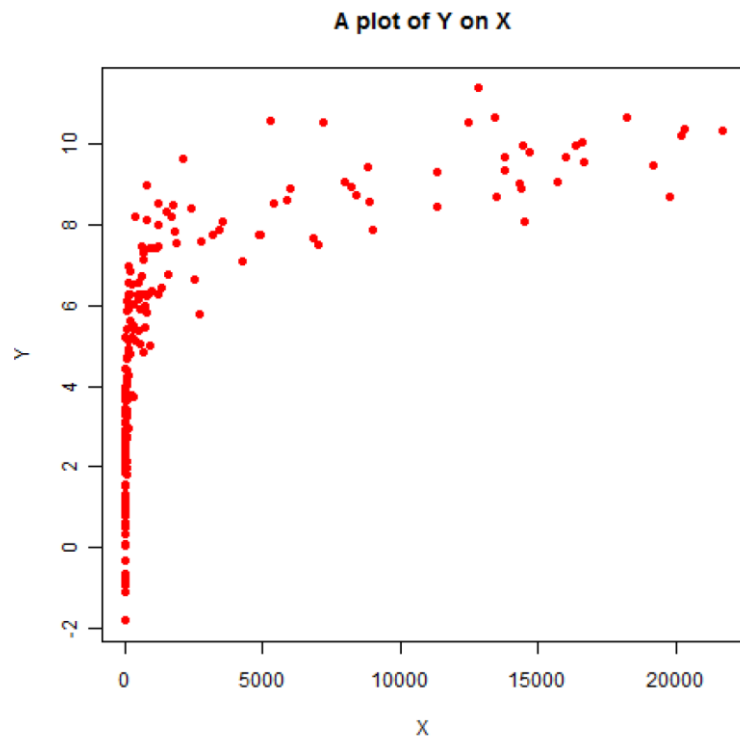
- .AIC is an asymptotic approximation
 - Number of params must be small compared to number of data points
- .True model must be in the parameterised family
- .Every model f in our family must map to a unique conditional probability distribution $p(\text{data}|f)$
- .Likelihood $L(f)$ function must be twice differentiable
- .Use it:
 - Linear regression, generalised linear models, constant bin width histogram estimation
- .Don't use it for
 - Multi-layer neural networks (uniqueness of p)
 - Mixture models
 - The uniform distribution (differentiability)
- .More details, e.g.:
- .http://www.csse.monash.edu.au/~dschmidt/ModelSectionTutorial1_SchmidtMakalic_2008.pdf

A more pragmatic strategy ...

Nonlinear Data, Transformations

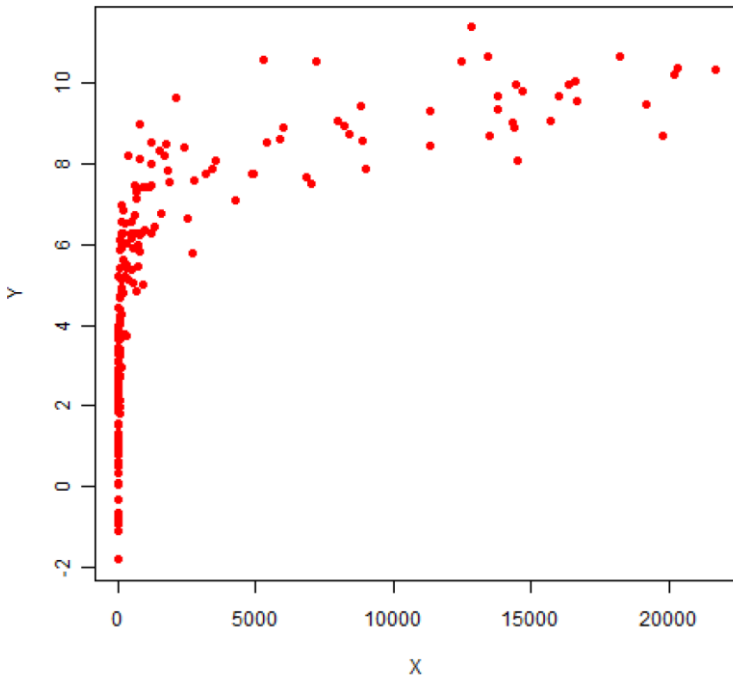
- Don't work with the raw data for X but with some transformation $f(X)$ which will hopefully be closer to linear
 - This is a bit like you were finding you were measuring the wrong thing, e.g., surface area might be a more direct measure of wind exposure than height
- How to know which transformation to use?
 - Sometimes there is a theoretical reason
 - Some relationships on a scatterplot may look familiar and suggest a fix
 - Can always experiment with different functions and assess them with R^2 or AIC

Example



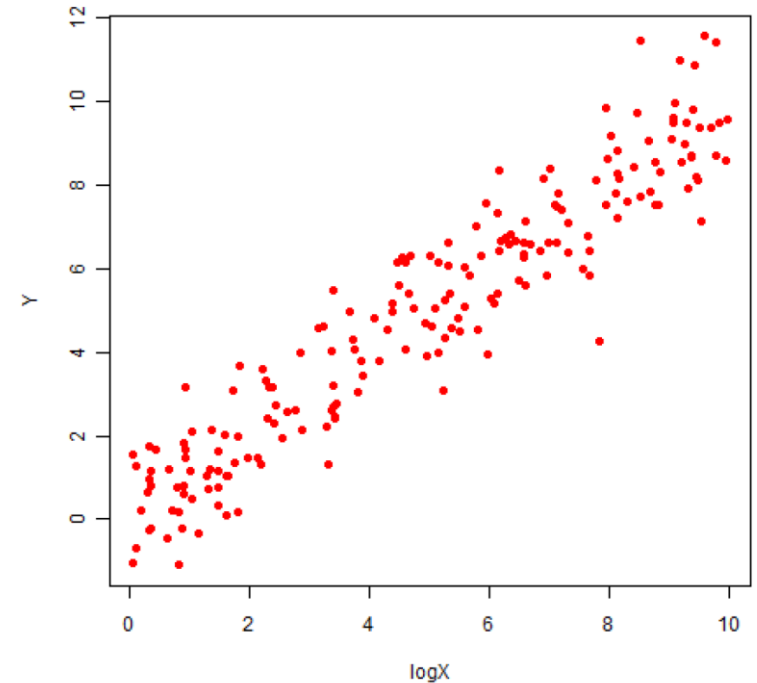
Example

A plot of Y on X



$$f(X)=\log(X)$$

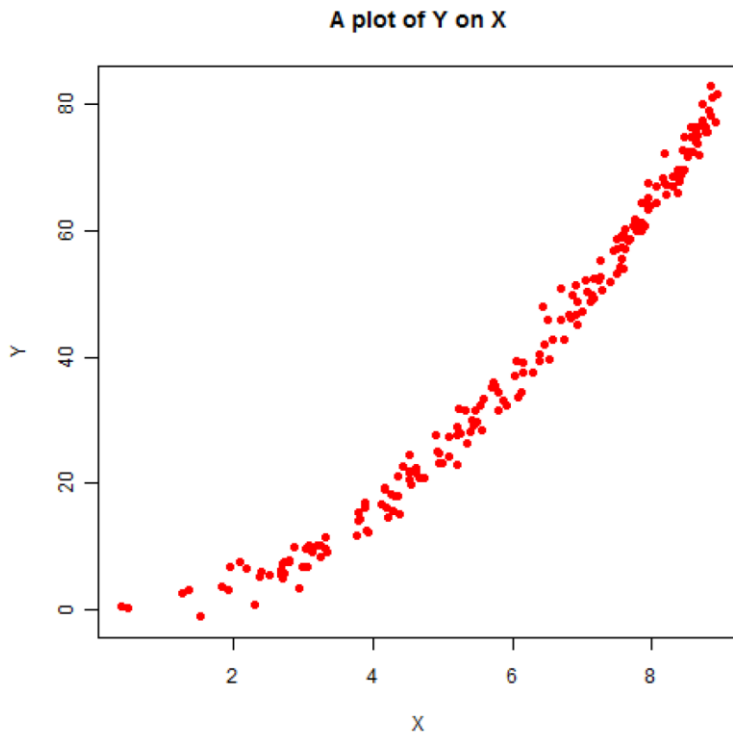
A plot of Y on log(X)



Correlation coefficient “improved” from 0.65 to 0.95

If relationship with transformed variables “looks more linear”, we will get a better fit with linear regression.

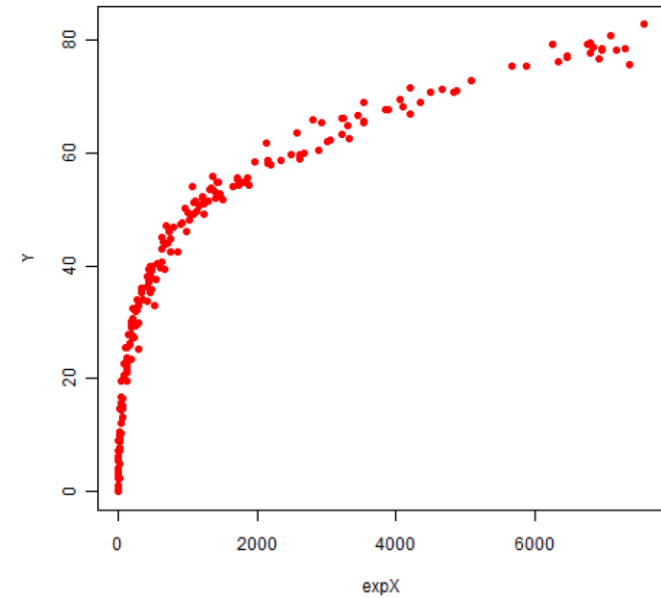
Example (2)



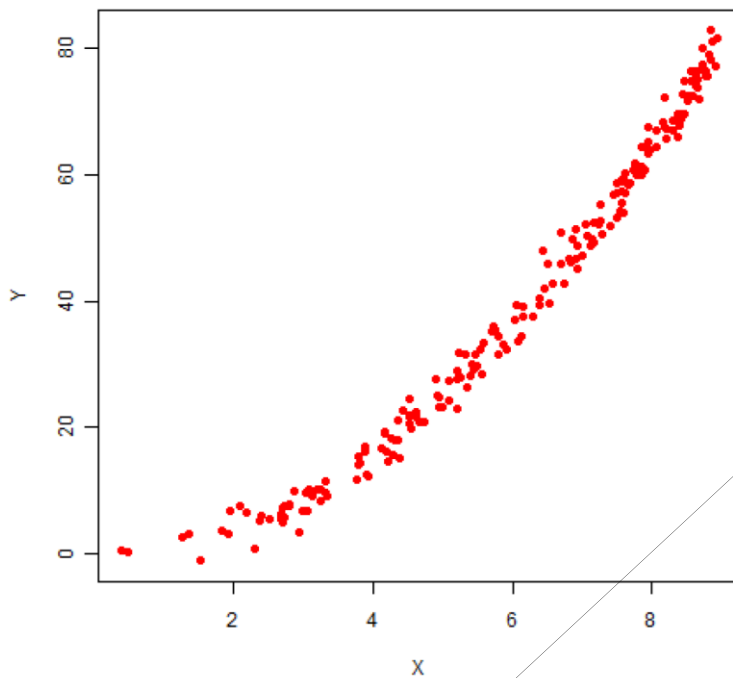
This time Y increases much faster than X.

Example (2)

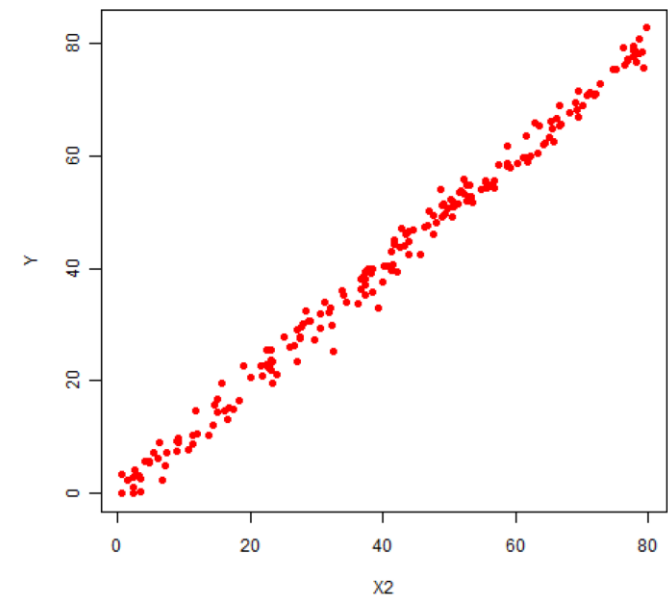
A plot of Y on exp(X)



A plot of Y on X



A plot of Y on X-squared



This time Y increases much faster than X.
Could try ...

$\exp(X)$?

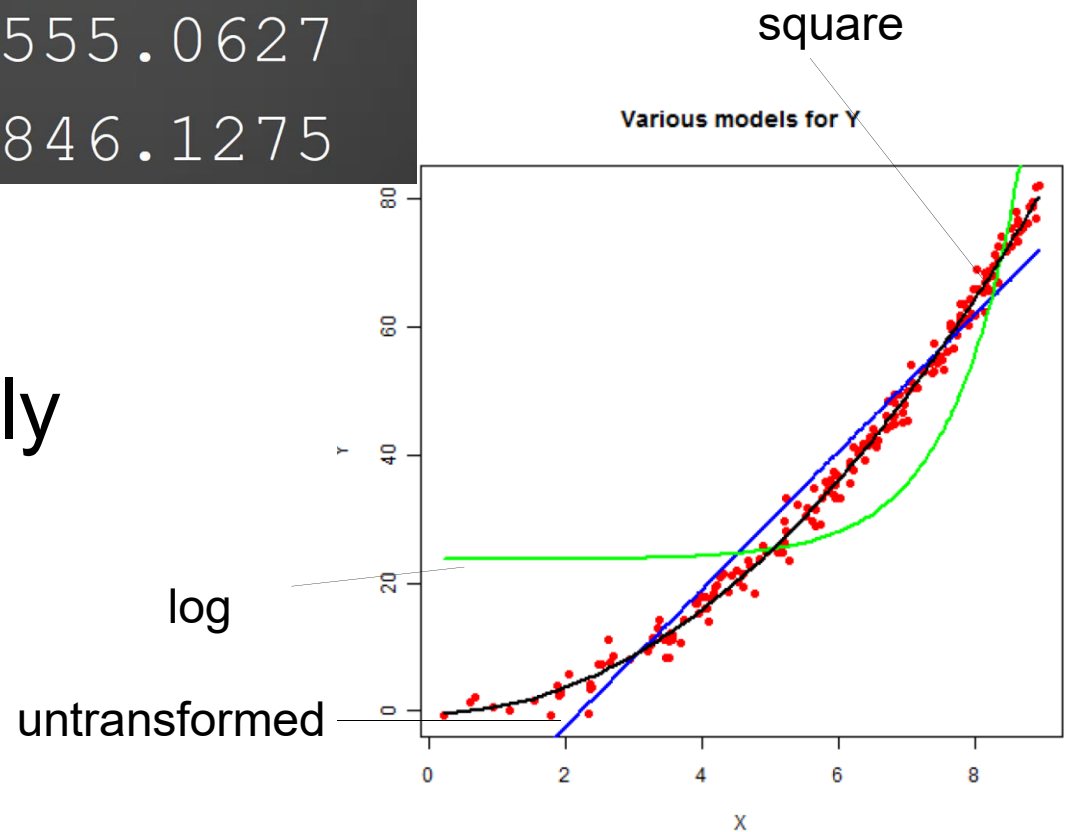
X^2 ?

Example, cont.

• Can fit regression models for all three cases and then use AIC to assess ...

	df	AIC
unchanged	3	1227.2688
expTrans	3	1555.0627
squareTrans	3	846.1275

• But this is also visually
• apparent ...



Dependent Data

Interaction Terms

.Regression on multiple variables assumes that these variables are independent

- i.e. if a predictor variable affects the outcome variable then its effect is independent of all other predictor variables
- E.g. the linear relationship between X_1 and Y is supposed to hold whatever values all other variables $X_2, X_3, \dots$ assume
- E.g.: Want to predict happiness from length of marriage
 - . For men happiness increases with length of marriage
 - . For women it decreases
 - . The relationship may be linear, but it is not independent of sex

Interaction Terms (2)

.If two predictor variables influence the outcome in a way that is **not additive**, we need to include an interaction term in the model to capture this effect

How to do it in R?

```
model = lm( Y ~ X1 * X2 )
```

.Will regress Y on X1 and X2 including an interaction term X1 by X2

.Equivalent to

```
model = lm( Y ~ X1 + X2 + X1:X2 )
```

.For syntax for more complex situations see

<http://stat.ethz.ch/R-manual/R-patched/library/stats/html/formula.html>

When to include Interaction Terms

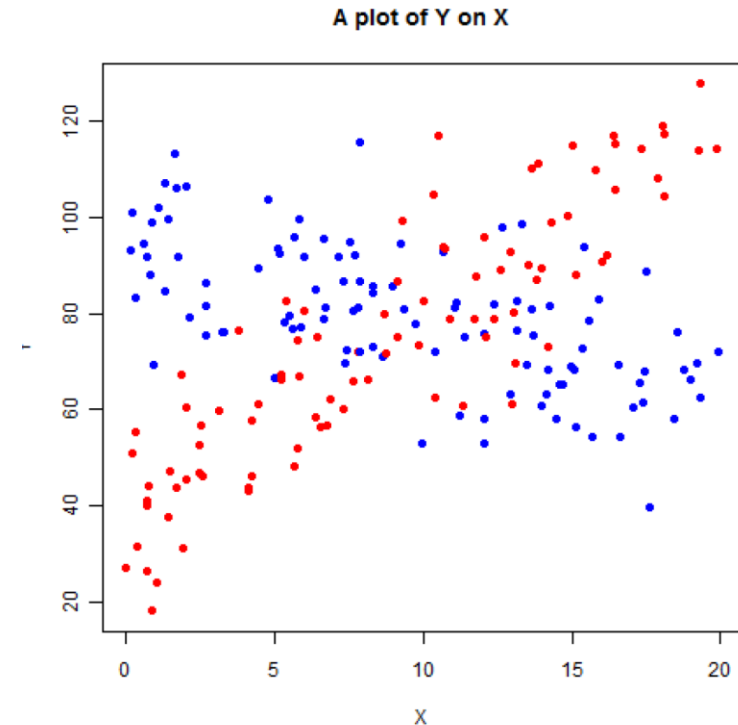
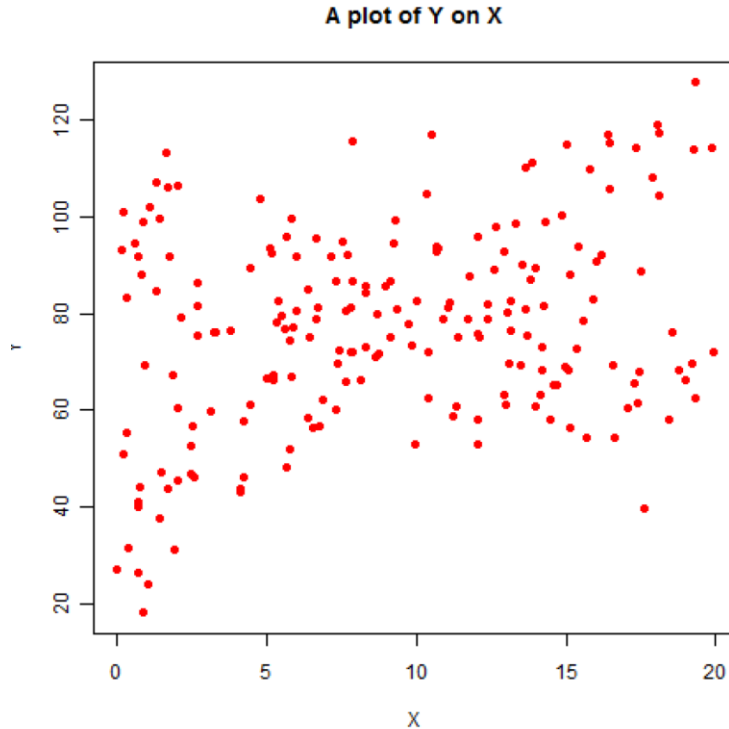
.If you have a large number of predictors, it is not practical to include all interactions

- Due to the combinatorics, you'll quickly have more parameters than data points ...
- Want to keep them to a minimum

.Include when:

- Theoretical reasons or direct questions that need to be answered; or suggested by other descriptive stats
- Once in there, up for elimination by model reduction

Example Interaction Terms



• The outcome measure Y (happiness) is dependent on the predictor X (length of marriage) but also on a categorical variable Group (**male** or **female**)

R practicalities

```
lm(formula = Y ~ X * Group)
```

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	33.4776	2.5465	13.15	<2e-16	***
X	4.3830	0.2144	20.44	<2e-16	***
GroupB	67.2934	3.2639	20.62	<2e-16	***
X:GroupB	-6.3115	0.2756	-22.91	<2e-16	***

```
---
```

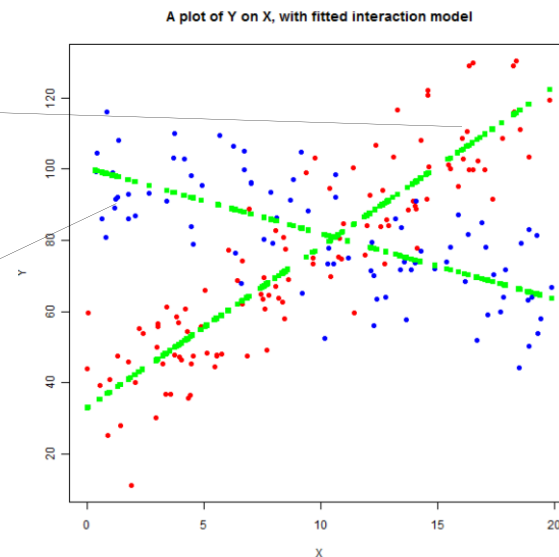
If you are in group A your prediction line is:

$$Y = 33.47 + 4.38X$$

If you are in group B your prediction line is:

$$Y = (33.47 + 67.29) + (4.38 - 6.31)X \text{ i.e.}$$

$$Y = 100.77 - 1.77X$$



R practicalities (2)

Can the model be reduced?

```
> drop1(fullModel)
Single term deletions

Model:
Y ~ X * Group

            Df Sum of Sq    RSS   AIC
<none>                 23674  962.76 —
X:Group     1      63366  87039 1221.16
```

AIC analysis confirms that this is not the case, i.e., our preferred model is the one with interactions.

Summary

•What you should remember:

- Aims of logistic regression
 - When to apply it.
- Logit transform and why it is used
- Interpreting (R) outputs of logistic regression
- How it actually works.
- You should be able to use it, have a play with the R example.
- Ideas how to deal with non-linear and dependent data
- Model reduction, AIC

Some Problems

• If you want to experiment a bit more with predicting movie success using linear/logistic regression models, explore the movie dataset I used in the lectures:

<https://www.southampton.ac.uk/~mb1a10/stats/filmData.txt>

• I can also recommend two good step-by-step tutorials:

- <https://www.r-bloggers.com/predicting-movie-ratings-with-imdb-data-and-r/>
- <https://rpubs.com/DocOfi/223687>

Some Problems

.A data set has been collected to relate the age of a learner to the outcome of driving tests. Carrying out logistic regression, somebody obtains a slope of $w=0.01$ and an intercept of $b=0.1$. What are the chances of a 100 years old applicant to pass the test?

Some Problems

- Somebody collects a data set to analyze examination outcomes (discriminating between fail, pass, and repeat) of students on a three-year BSc degree and carries out multinomial logistic regression to predict the outcome dependent on the year of study. Results give:
 - intercept (fail)=1 slope fail=-1
 - intercept (pass)=3 slope (pass)=-1/2
- What is the chance of a student having to repeat the 3rd year?